

Identifying Hidden Graphs with Quantum Walks

From BQP-Completeness to an Exponential Separation

arXiv:2605.11228

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What I work on these days

- ▶ quantum computing needs **new algorithms**, and **new applications** for the ones we have
- ▶ the **Genesis Mission** is aimed at exactly that: using AI for quantum algorithm discovery
- ▶ $\Rightarrow$ the **paper** behind this talk, [arXiv:2605.11228](https://arxiv.org/abs/2605.11228), was a warm-up for it
 - ▶ the ideas are mine, found without AI
 - ▶ the **numerics** took non-trivial AI assistance

The problem

We are handed a Hamiltonian H , together with a start state $|\psi\rangle$.

- ▶ H **sparse** and **exponentially large**; $|\psi\rangle$ a **computational basis state**
- ▶ H and $|\psi\rangle$ define the **signal**:

$$f(t) := \langle \psi | e^{-iHt} | \psi \rangle$$

- ▶ a **pair** of *small* graphs A_0, A_1 , and a vertex v , all known to us
- ▶ we are **promised** that H acts as A_b on the subspace the evolution explores, for one of $b \in \{0, 1\}$

Decide b from the signal.

The promise in detail: H on the Krylov subspace

- ▶ the **Krylov subspace** $\mathcal{K} = \text{span}\{H^k|\psi\rangle\}$: the smallest subspace containing $|\psi\rangle$ that the evolution never leaves
- ▶ the **promise** is that $\mathcal{K}$ is **tiny**, and that on it H is a graph:

$$|\psi\rangle \leftrightarrow |v\rangle, \quad H|_{\mathcal{K}} = \underbrace{A_b}_{\text{adjacency matrix, small graph}}$$

- ▶ the correspondence runs through every power, $H^k|\psi\rangle \leftrightarrow A_b^k|v\rangle$
 $\Rightarrow$ the signal transfers:

$$f(t) = \langle\psi|e^{-iHt}|\psi\rangle = \langle v|e^{-iA_b t}|v\rangle$$

A quantum walk: $e^{-iA_b t}$, on a graph, from a vertex.

An example: one flipped spin on a ring

- ▶ N spins on a ring, nearest-neighbour hopping, 2^N dimensions:

$$H = \sum_j (\sigma_j^+ \sigma_{j+1}^- + \sigma_j^- \sigma_{j+1}^+) \quad (2\text{-local})$$

- ▶ start from **one flipped spin**; write $|j\rangle$ for the flip at site j , a computational basis state
- ▶ $H|j\rangle = |j-1\rangle + |j+1\rangle$: the flip hops, and no others appear
- ▶ $\Rightarrow \mathcal{K}$ is the N positions, and $H|_{\mathcal{K}} = A(C_N)$

**2^N -dimensional local Hamiltonian;
on the Krylov subspace, the N -cycle.**

How is H given?

- ▶ the algorithm only ever *uses* the signal $f(t) = \langle \psi | e^{-iHt} | \psi \rangle$
- ▶ to observe f , it has to *simulate* e^{-iHt}
 $\Rightarrow H$ has to be specified somehow
- ▶ **as local terms**: a sum of $O(1)$ -local terms, written down explicitly; the example above, and the BQP instance
- ▶ **as a sparse black box**: one query returns the nonzero entries of one row; the spired instance
- ▶ both are enough to simulate e^{-iHt} efficiently

Local terms, or one row at a time.

A simple quantum algorithm; classically hard

- ▶ A_0, A_1 are known $\Rightarrow$ both **candidate signals** are computable, $f_b(t) = \langle v | e^{-iA_b t} | v \rangle$ for $b \in \{0, 1\}$, and f is one of them
- ▶ find the time t^* at which they differ the most
- ▶ run the evolution at t^* , and see which value comes back
- ▶ the small graphs give both: the time t^* , and the two values $f_0(t^*)$ and $f_1(t^*)$

Quantum side: easy

- ▶ controlled e^{-iHt^*} , with a Hadamard test to estimate $f(t^*)$

Classical side: hard

- ▶ no generic classical way to evaluate a signal living in 2^N dimensions

The algorithm: the classical half

- ▶ **before the machine is on:** A_0 and A_1 are small, so this is ordinary linear algebra

t^* = the time at which f_0 and f_1 differ the most

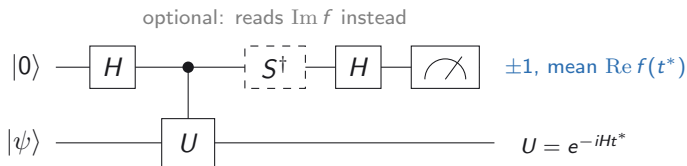
$$\text{Dis} = |f_0(t^*) - f_1(t^*)|$$

- ▶ no sampling: t^* and Dis are fixed numbers, computed once

Choosing t^* costs no quantum computer at all.

The algorithm: the quantum half

- ▶ **at $t = t^*$:** a **Hadamard test**, one controlled e^{-iHt^*} per repetition
- ▶ $N_{\text{rep}} = \mathcal{O}(1/\text{Dis}^2)$ of them



Reading the signal needs a quantum interferometer.

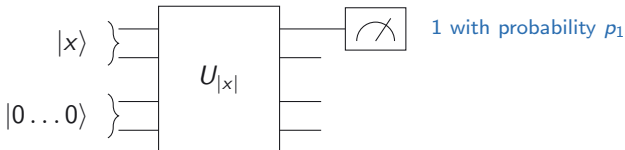
Two instances

	big Hamiltonian H	small graph A_b
BQP	clock $\otimes$ circuit	C_m vs. C_{2m}
spired graphs	spired prism / Möbius G_{spire}	towered graph G_{tower}

- ▶ **BQP** — deciding whether C_m or C_{2m} hides inside the Hamiltonian captures the *full power of quantum computation*. A **theorem**.
- ▶ **spired graphs** — deciding whether a *prism* or a *Möbius ladder* hides inside the big obfuscated graph is **conjectured** to be classically hard, in the **black-box model**.

Formalizing the power of quantum computing

A language $L \subseteq \{0, 1\}^*$: given an input x , decide whether $x \in L$.



- ▶ the circuit $U_{|x|}$ is built from $\text{poly}(|x|)$ elementary gates, on N qubits: the input x , plus ancillas
- ▶ one **output qubit** is measured $\Rightarrow$ outcome 1 with probability p_1
- ▶ **promise:** $x \in L \Rightarrow p_1 > 2/3$, $x \notin L \Rightarrow p_1 < 1/3$

That is BQP. One bit: accept or reject.

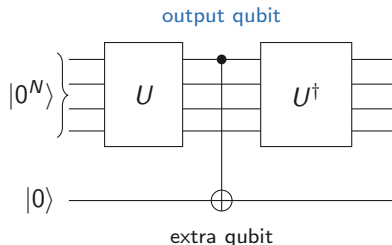
Two simplifications, both for free

- ▶ **Absorb the input.** Flip the qubits where x has a 1
 $\Rightarrow$ x is part of the circuit; the start is always $|0^N\rangle$
- ▶ **Amplify the gap.** Repeat and take the majority
 $\Rightarrow$ w.l.o.g. $p_1 = 0$ (**NO**) or $p_1 = 1$ (**YES**)

One circuit U , start $|0^N\rangle$, answer one bit b .

Step 1: from a circuit to an involution

- ▶ adjoin **one extra qubit**; the circuit copies its answer onto it, then undoes itself
- ▶ start from $|\text{init}\rangle = |0^N\rangle|0\rangle$



$$V = (U^\dagger \otimes I) \text{CNOT} (U \otimes I)$$

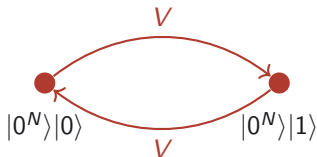
A Hermitian involution: $V = V^\dagger, \quad V^2 = I$

The answer is the dimension

- $V^2 = I \Rightarrow$ the orbit of $|\text{init}\rangle$ is **one point or two**



NO ($p_1 = 0$): $\dim = 1$



YES ($p_1 = 1$): $\dim = 2$

- in the YES case the extra qubit flips, so V **swaps** two states instead of fixing one

Step 2: from the involution to a local Hamiltonian

- ▶ factor $V = V_m \cdots V_1$ into m elementary gates, add a clock register of dimension m
- ▶ define the **forward-time operator**:

$$F = \sum_{t=0}^{m-1} |t+1 \bmod m\rangle\langle t|_c \otimes V_{t+1}$$
$$H = F + F^\dagger$$

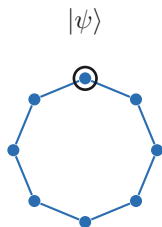
- ▶ F walks on the clock, a cycle of length m ; each step applies one gate to the computational register
- ▶ H is $O(1)$ -**local**, with $\text{Spec}(H) \subseteq [-2, 2]$
- ▶ start from $|\psi\rangle = |0\rangle_c \otimes |\text{init}\rangle$

H has dimension $m \cdot 2^{N+1}$.

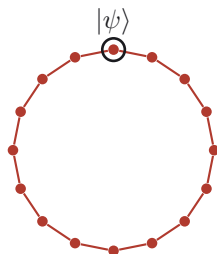
The two cycles

One lap of the clock applies the whole of V :

$$F^m|\psi\rangle = |0\rangle_c \otimes V|\text{init}\rangle$$



NO: C_m
one lap comes back



YES: C_{2m}
two laps to come back

The time evolution is a walk on a cycle

- ▶ H acts on $m \cdot 2^{N+1}$ dimensions
- ▶ but from $|\psi\rangle$ the evolution never leaves the orbit of F :
 - ▶ m states in the **NO** case
 - ▶ $2m$ states in the **YES** case
- ▶ the orbit is orthonormal and H -invariant
- ▶ each vertex is the circuit after k gates: $|k\rangle_c \otimes V_k \cdots V_1 |\text{init}\rangle$
- ▶ $\Rightarrow$ one step along the cycle is **one gate**

Walking the cycle is running the circuit.

The signal, winding by winding

- ▶ the walk on an **infinite line** is a Bessel function:

$$\langle x | e^{-iAt} | 0 \rangle = (-i)^x J_x(2t) \quad [\text{Jacobi-Anger}]$$

- ▶ a cycle is that line **wrapped** $\Rightarrow$ sum over the images, one per **winding** $x = jn$:

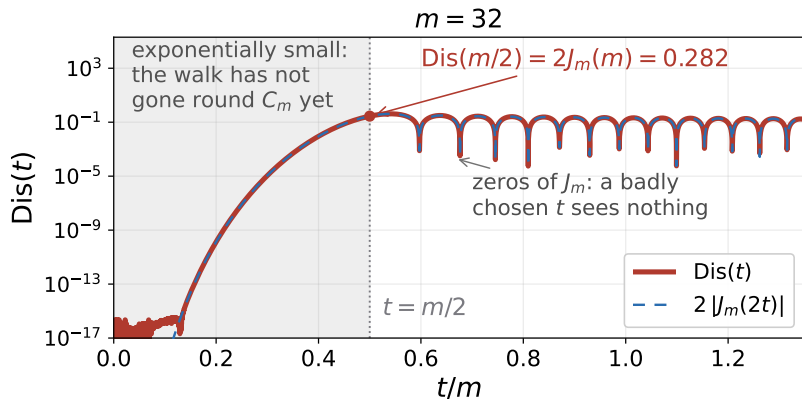
$$n = 2^b m, \quad b \in \{0, 1\} \quad f_b(t) = \sum_{j \in \mathbb{Z}} (-i)^{jn} J_{jn}(2t)$$

- ▶ the **even** windings are shared, and cancel:

$$f_0(t) - f_1(t) = \sum_{j \text{ odd}} (-i)^{jm} J_{jm}(2t) \approx 2(-i)^m J_m(2t)$$

One Bessel function, to an excellent approximation.

What the difference looks like



$\text{Dis}(t) = |f_0(t) - f_1(t)|$ at $m = 32$, log scale. Dashed: the first odd winding, $2|J_m(2t)|$.

Exponentially small until the walk has gone once round C_m ; then the first odd winding.

When to look, and how many times

- ▶ the amplitude at distance x is $J_x(2t)$, and J_x is exponentially small until its argument reaches its order
- ▶ $\Rightarrow$ **light cone**: distance $2t$ by time t
- ▶ at $t = m/2$: **once round** C_m , only **half way** round C_{2m}

$$t^* = \frac{m}{2}, \quad \text{Dis} = 2 J_m(m) = \Theta(m^{-1/3})$$

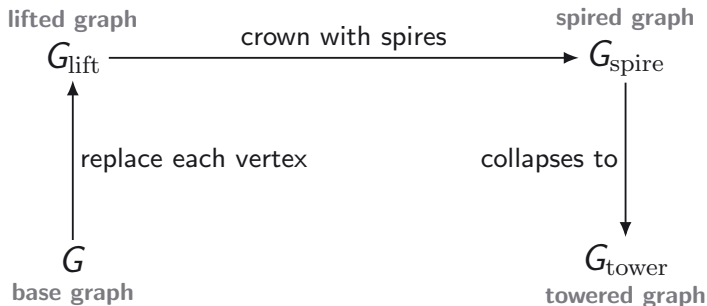
- ▶ $\Rightarrow \mathcal{O}(m^{2/3})$ repetitions, polynomial

In BQP we did not choose the graph

- ▶ one natural reduction: a circuit $\Rightarrow$ a clock Hamiltonian $\Rightarrow$ a walk on C_m or C_{2m}

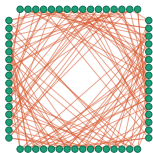
Now hide a graph we choose, inside a graph.

Dramatis personae

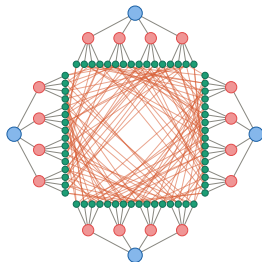


- pick *any* regular G on n vertices $\Rightarrow G_{\text{spire}}$ is complicated, and G **determines its symmetry**; the symmetry is what will collapse the walk

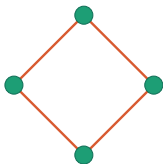
Dramatis personae



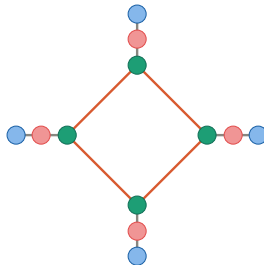
lifted graph



spired graph

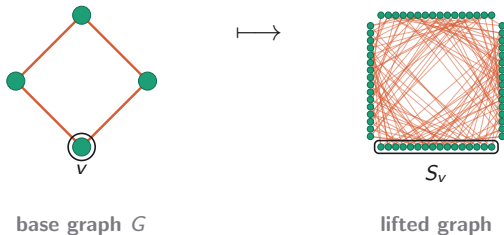


base graph



towered graph

The lift



- ▶ every vertex becomes a **cluster** of look-alikes; every edge of G becomes a **bipartite graph** between the two clusters
- ▶ **regular** — the same number of partners for everyone $\Rightarrow$ the walk **never leaves** the uniform cluster states
- ▶ **random**, and every label **shuffled** $\Rightarrow$ every vertex looks alike: degree D , and no landmark anywhere

Regular buys the collapse. Random buys the hiding.

But now we are stuck too

- ▶ after the lift, *being at v* means the uniform state over v 's cluster:

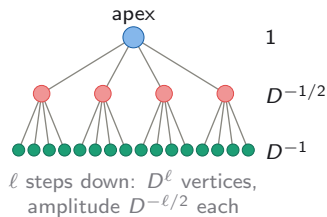
$$|S_v\rangle = |S_v|^{-1/2} \sum_{x \in S_v} |x\rangle$$

- ▶ the graph is reached only through an **oracle**, and every label is a **random string**; we cannot list a cluster's members
- ▶ $\Rightarrow$ it cannot hand us $|S_v\rangle$

To walk on G we need a state we cannot prepare.

The spire is the way in

- ▶ crown each cluster with a **tree** of height L : one **apex** on top
- ▶ the apex is a *single label*, which the oracle *can* hand over
- ▶ $\Rightarrow$ L steps down land exactly on $|S_v\rangle$



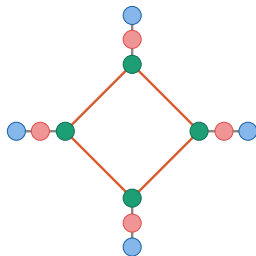
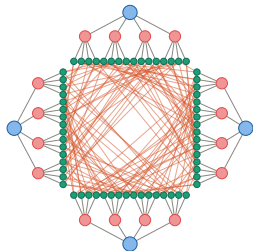
The spire turns the one label the oracle can give into the state we could not prepare.

Two collapses

- ▶ the walk never leaves the span of the **level states**: every level, not just the cluster
- ▶ $\Rightarrow$ each **spire** becomes a **tower**: tree $\longrightarrow$ path
- ▶ $\Rightarrow$ each **cluster** becomes a **vertex**: D^L look-alikes $\longrightarrow$ one

G_{tower} is the quotient of G_{spire} under the level-set partition.

An exponentially large walk, run by an n^2 matrix



G_{spire} : $\Theta(nD^L)$ vertices, *hidden* G_{tower} : $n(L+1) = n^2$ vertices, $L = n-1$, *on paper*

$$H|_{\mathcal{K}} = A_{\text{tower}}$$

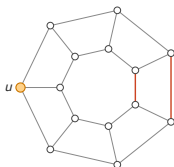
The obfuscating oracle

- ▶ **in**: a label $\Rightarrow$ **out**: the labels of its neighbours, and **nothing** else ($\perp$ if it is not a label); the **same** oracle serves the classical algorithm
- ▶ labels are random a -bit strings, $a = 2 \log_2 |V_{\text{spire}}| \Rightarrow$ a random string is a label with probability $\leq 1/|V_{\text{spire}}|$
- ▶ we are given **one** label: one apex, our entry point

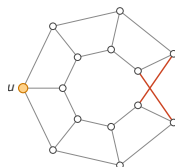
You cannot guess your way in. You can only walk.

Now we choose what to hide

All of that works for *any* regular G . So take the **hardest** pair.



prism Y_7



Möbius ladder M_7

- ▶ 3-regular on $n = 2m$ vertices: a ladder closed straight, or twisted
- ▶ two red edges in each: the *entire* difference
- ▶ u : our entry point, kept far from both

One oracle, two candidates

- ▶ **given:** H , through the oracle only; $\Theta(nD^L)$ vertices, nothing else
- ▶ **start:** the apex of one spire, a single label
- ▶ **promised:** on the subspace the walk explores, H acts as A_{tower} , built either from the **prism** or from the **Möbius ladder**
- ▶ $\Rightarrow$ decide which

Everything that remains is classical: two $n^2 \times n^2$ matrices, and where their signals differ most.

What the towers do to the spectrum

- ▶ we need both candidate signals, $f(t) = \sum_{\lambda} w_{\lambda} e^{-i\lambda t}$ over the spectrum of A_{tower}
- ▶ $n^2 \times n^2 \Rightarrow \mathcal{O}(n^6)$: hopeless at $n \approx 10^4$
- ▶ but the towers give A_{tower} a structure, and a symmetry:

$$A_{\text{tower}} = A \otimes |L\rangle\langle L| + I \otimes P \qquad [A_{\text{tower}}, A \otimes I] = 0$$

So A_{tower} splits into one block per eigenvalue of the hidden graph.

A direct sum of tridiagonal matrices

$$A_{\text{tower}} \cong \bigoplus_{j=1}^n T^{(\mu_j)}, \quad T^{(\mu)} = \mu |L\rangle\langle L| + P$$

$\mu_1, \dots, \mu_n$: the eigenvalues of the hidden graph. Each block is

tridiagonal: the path with weight γ , and a single diagonal entry μ at the bottom.

we start here (the apex)

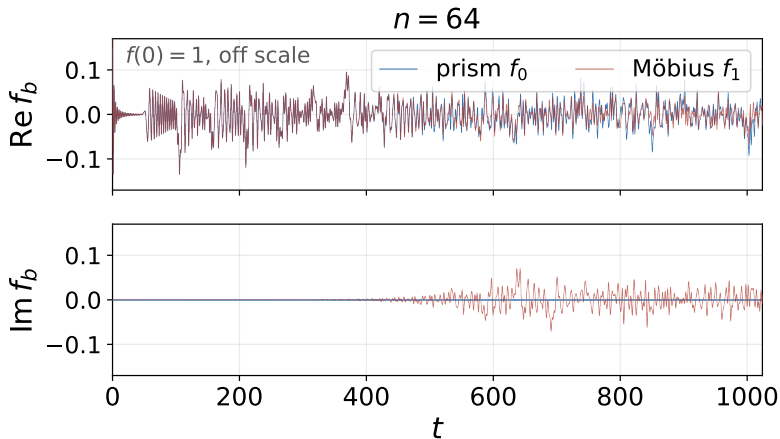
the hidden graph acts here



- ▶ eigenvalues and eigenvectors from a **Chebyshev recurrence**
- ▶ n **independent** scalar problems $\Rightarrow$ the whole signal in $\tilde{O}(n^2)$

The boundary potential is the hidden graph's eigenvalue.

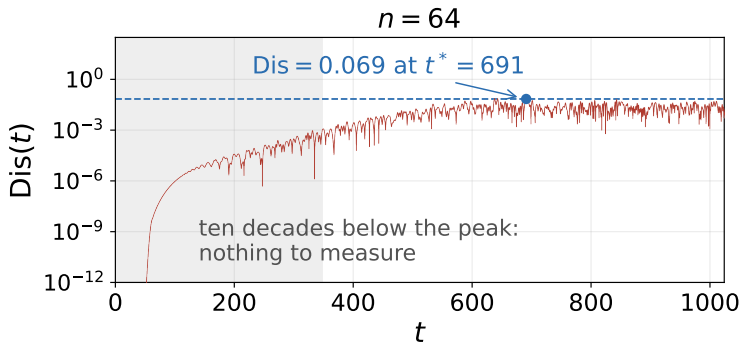
The two signals



Both signals decay to a band of size $\Theta(1/n)$ and stay there.

The real parts are one band; the imaginary parts are not.

Their difference



$\text{Dis}(t)$, log scale: exponentially small until the walk reaches the two edges the candidates differ in, then a band whose peak is $\text{Dis} = 0.069$ against a typical 0.027.

The peak stands $2.6\times$ above the typical level.

Why there is a peak, and what it costs

- ▶ $f_0 - f_1$ is about n^2 phases of size about $1/n^2 \Rightarrow$ Dis is typically $1/n$ at a random time
- ▶ at t^* enough of them align to stand $\sqrt{\log n}$ above that level
 $\Rightarrow \text{Dis} = \Theta(\sqrt{\log n} / n)$ [NUMERICS, $n \leq 10^4$]
- ▶ and t^* is not just somewhere inside the horizon: it sits at $0.16 n^2$ at every size tested

quantum $\sim n^2$ repetitions, $\tilde{O}(n^4)$ gates [CONJ]

classical exponentially many queries [CONJ]

The spires give us the cluster superposition, and with it the collapse; they are also why t^* is computed and not derived.

What is proved, and what is not

- ▶ **The mechanism is proved**
 - ▶ the walk really is C_m or C_{2m}
 - ▶ the collapse $H|_{\mathcal{K}} = A_{\text{tower}}$
 - ▶ the spectral theory of G_{tower}
- ▶ **The separation is not:** two *conjectures*
 - ▶ classical hardness, *relative to the oracle*
 - ▶ the total quantum cost, $\tilde{O}(n^4)$ gates

Hidden subgroup, welded trees, and this one

- ▶ **hidden subgroup** — *which one is it?* answered with a group to exploit (discrete log, Shor)
- ▶ **welded trees** — *can you get there?* no group, and a proved lower bound
 - ▶ G_{spire} specializes to it at $G = K_2$
- ▶ **here** — *which one is it?* with no group to exploit

All of it is relative to the oracle.

Open:

- ▶ **exit-finding**: from the seed apex, find the *other* apices; the bridge between the two questions
- ▶ what else can play the **spire**'s part: funnel the amplitude in, look like nothing from outside
- ▶ find this structure arising *on its own*, somewhere in mathematics

Where this comes from

- ▶ **this talk** — *Quantum algorithm for identifying hidden graphs*, arXiv:2605.11228
- ▶ **BQP-complete problems** — Janzing & Wocjan, *Theory of Computing* 2007, *QIC* 2010
- ▶ **spectral graph theory** — with C. Elphick, 2013–2019
- ▶ **welded trees** — Childs et al., *STOC* 2003

The BQP construction is not part of the spired graph paper; this is the context it sits in.